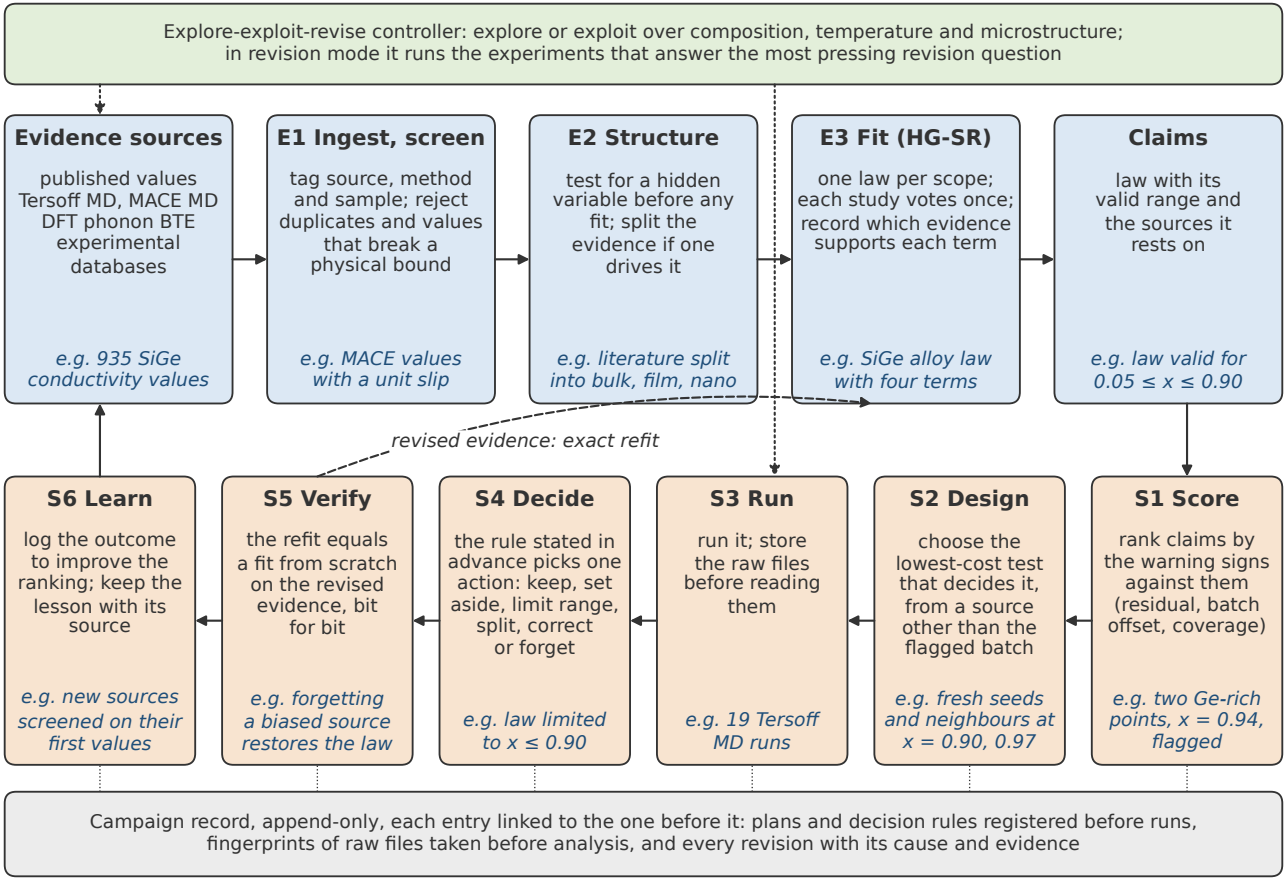
Explore-exploit-revise controller: explore or exploit over composition, temperature and microstructure; in revision mode it runs the experiments that answer the most pressing revision question
Evidence sources
published values Tersoff MD, MACE MD DFT phonon BTE experimental databases
e.g. 935 SiGe conductivity values
E1 Ingest, screen
tag source, method and sample; reject duplicates and values that break a physical bound
e.g. MACE values with a unit slip
E2 Structure
test for a hidden variable before any fit; split the evidence if one drives it
e.g. literature split into bulk, film, nano
E3 Fit (HG-SR)
one law per scope; each study votes once; record which evidence supports each term
e.g. SiGe alloy law with four terms
Claims
law with its valid range and the sources it rests on
e.g. law valid for 0.05 ≤ x ≤ 0.90
revised evidence: exact refit
S6 Learn
log the outcome to improve the ranking; keep the lesson with its source
e.g. new sources screened on their first values
S5 Verify
the refit equals a fit from scratch on the revised evidence, bit for bit
e.g. forgetting a biased source restores the law
S4 Decide
the rule stated in advance picks one action: keep, set aside, limit range, split, correct or forget
e.g. law limited to x ≤ 0.90
S3 Run
run it; store the raw files before reading them
e.g. 19 Tersoff MD runs
S2 Design
choose the lowest-cost test that decides it, from a source other than the flagged batch
e.g. fresh seeds and neighbours at x = 0.90, 0.97
S1 Score
rank claims by the warning signs against them (residual, batch offset, coverage)
e.g. two Ge-rich points, x = 0.94, flagged
Campaign record, append-only, each entry linked to the one before it: plans and decision rules registered before runs, fingerprints of raw files taken before analysis, and every revision with its cause and evidence